

AI-Based Information Extraction from High-Resolution Satellites: Building Extraction and Disaster Analysis

**Tsutomu Yamanokuchi, Kurodo Saito, Hiroki Mori, Yusuke Adachi, Kazuho Kayama,
Kai Kubo and Tetsuya Kaku**
Remote Sensing Technology Center of Japan

Company Outline - ABOUT RESTEC -

■ Remote Sensing Technology Center of Japan (RESTEC)

- ADDRESS : TOKYU REIT Toranomon Building 3F 3-17-1 Toranomon, Minato-ku, Tokyo 105-0001, Japan
- Established : **August 1st, 1975** 【more than 50years history】
- Total Employees : Approx. 300 (Incl. dispatched staff)

3 main offices

HQ :

Central of Tokyo

Tsukuba :

In JAXA Tsukuba space center

Hatoyama :

In JAXA Hatoyama

Earth observation center



Org. Chart

The RESTEC Vision

■ Paving the way for a prosperous future with remote sensing technology

- Enhancing, Connecting and Sharing -

Enhancing

RESTEC will continuously pursue the development and use of remote sensing technology to further enhance its technological capabilities and values.

Connecting

RESTEC will connect society, daily life, and humanity by making the outputs of the development and application of remote sensing technology into useful and valuable information in a variety of domains.

Sharing

RESTEC will share the outputs of the development and application of remote sensing technology with people all over the world.

6 Business field of RESTEC



■ Satellite Earth Observation



■ Research and Development



■ Think tank



■ Solution



■ Capacity building



■ Enhancement of Utilization

RESTEC's challenge to EO satellite data with AI

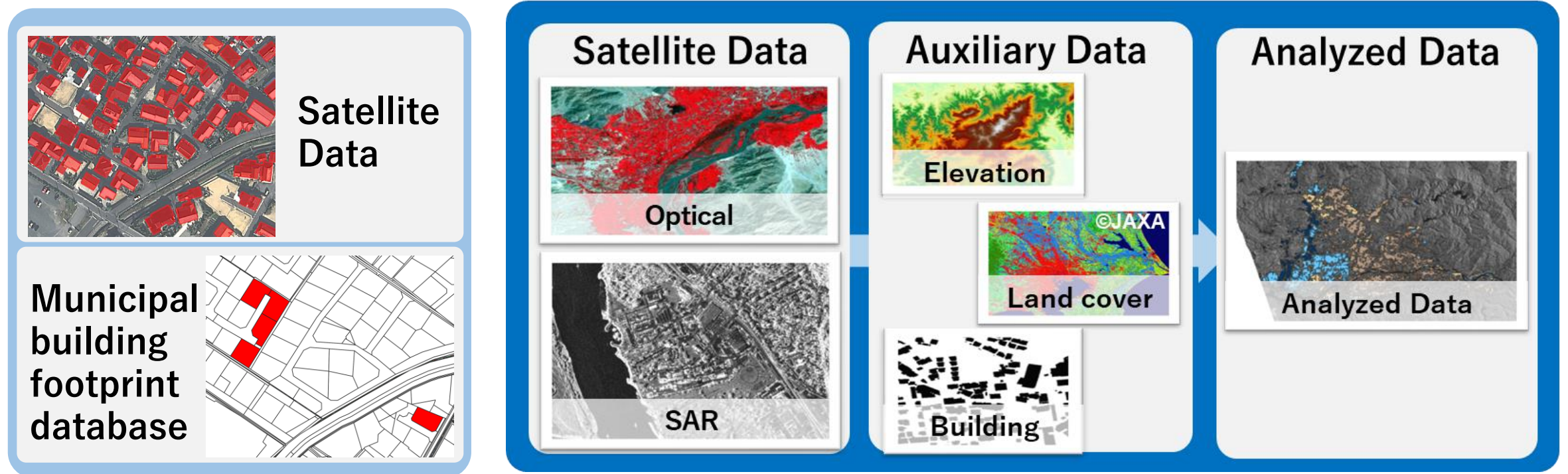
- RESTEC have been developing solutions for various field of satellite data utilization.

1. **Extraction of individual houses**

→Technologies for integration of segmentation-based AI

2. **Multi-Source Change Detection for disaster analysis**

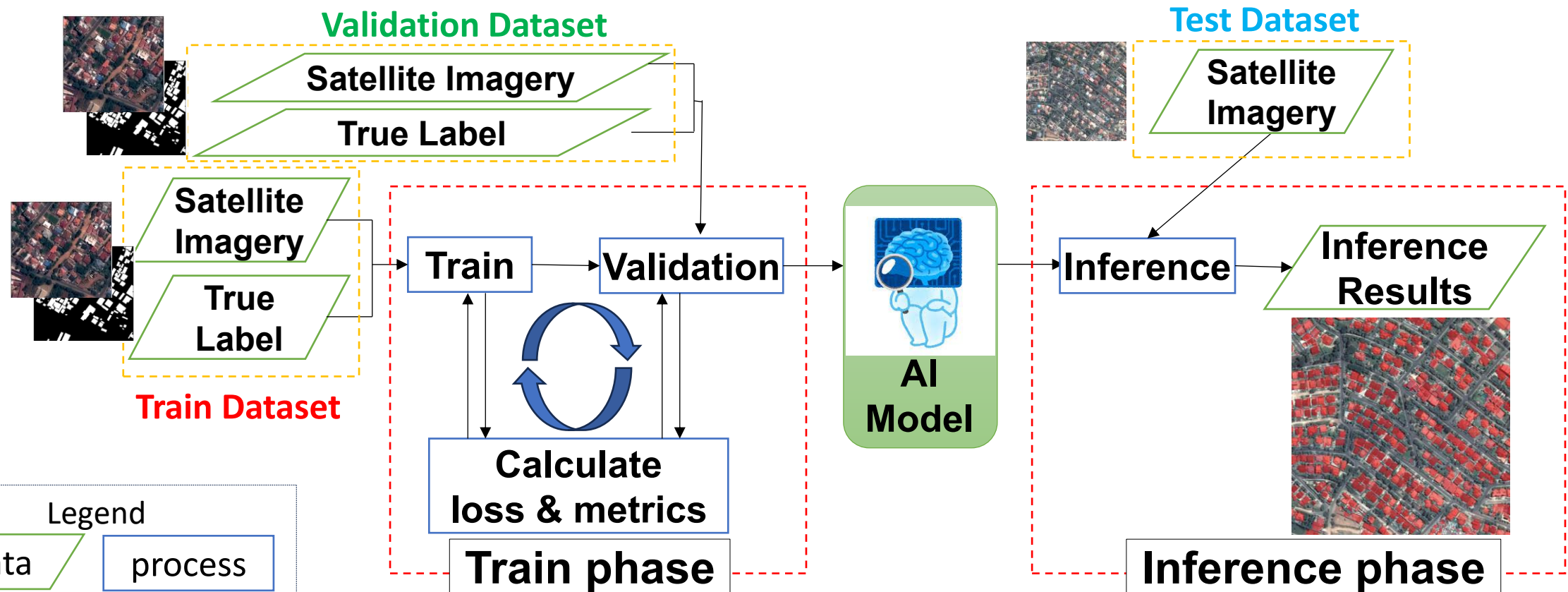
→Technologies for more effectively and quickly delivering information to users



Extraction of Individual houses

Workflow of Building Extraction AI Using Deep Learning

- Deep learning requires repeated training to improve accuracy.
- Train the model using “**Training dataset**”, validate the accuracy of the trained model with “**Validation data**” and finalize the model with the best accuracy.
- The trained model is applied using “**Test Dataset**” that were not used for training.
- All processes were implemented in **Python using PyTorch**.



Training Data for Building Extraction AI

The quality and characteristics of training data strongly affect model accuracy.

① Open benchmark datasets

e.g., WHU Building Dataset, SpaceNet datasets

- 0.3m Aerial / Satellite images
- **High-quality images & labels,**
but limited regions.

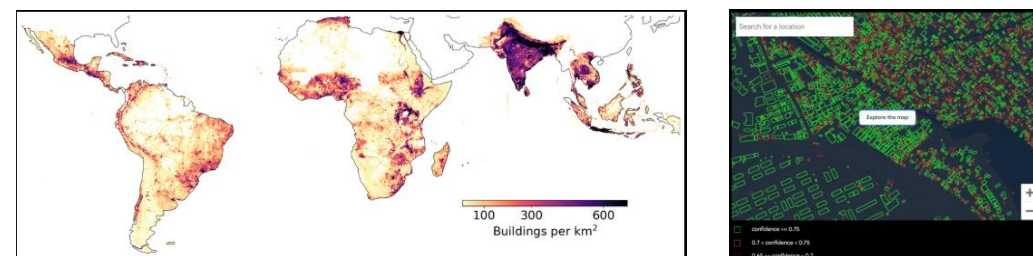


WHU Building Dataset

② Global building polygon dataset

e.g., Google Open Buildings

- extracted from high-resolution satellite imagery (50cm Pleiades, WV02...)
- **Large-scale coverage,**
Large-scale coverage, but building footprints may not accurately match the target imagery.

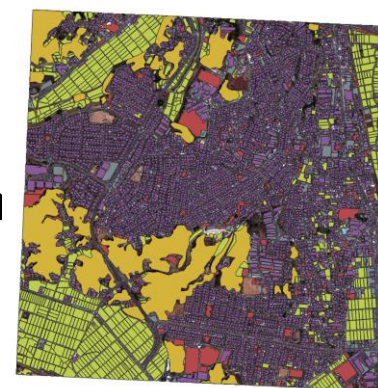


Global building polygon

③ Custom annotation data

Created by outsourcing high-quality building annotation

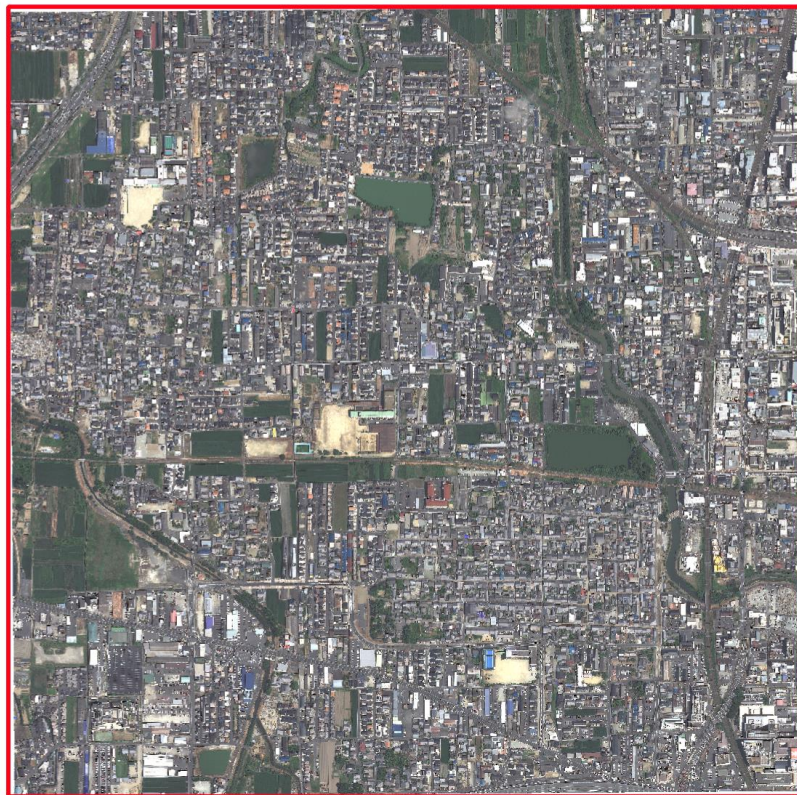
- **High accuracy and project-specific,**
but requires cost and time.



- ☒ WV02_20220309_0054
- ☒ fude_WV02_20220309 — fude_wv02_20220309
- ☒ polygon_WV02_20220309_SolarPanel — template
- ☒ polygon_WV02_20220309_Parking — template
- ☒ polygon_WV02_20220309_Forest — template
- ☒ polygon_WV02_20220309_Building — template
- ☒ polygon_WV02_20220309_BareLand — template
- ☒ img_WV02_20220309_0054_TOA

Example of Outsourced Annotation Polygons

Overview of Analysis Area



Approx. 1.6km

← Approx. 1.6km →

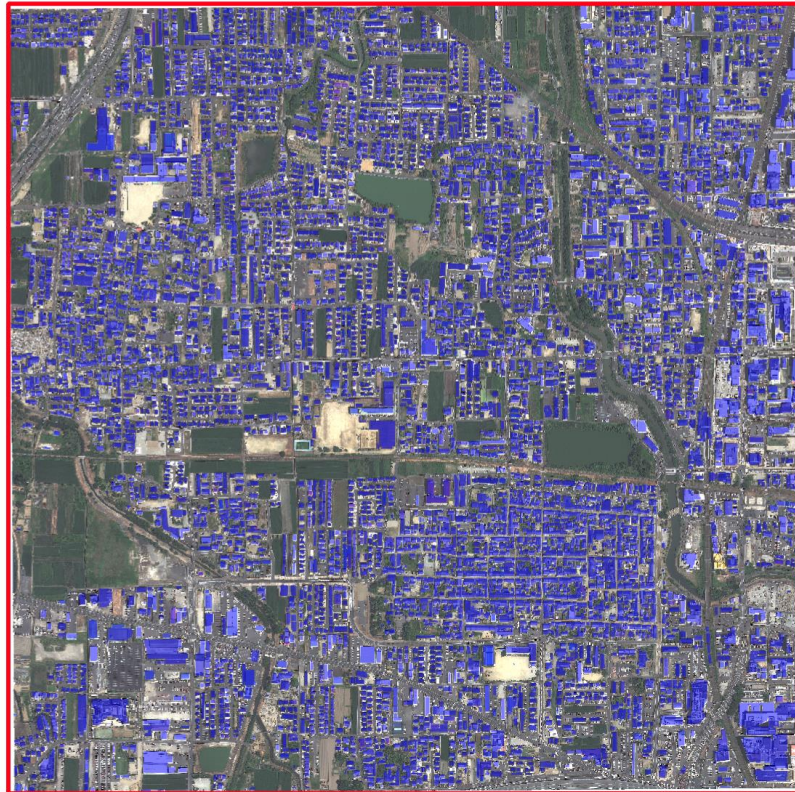
Pleiades Neo 2023/8 10:48
Incidence angle: 8.9°



Pleiades Neo 2025/1 10:48
Incidence angle: 7.6°

Overview of Analysis Area

- : 1st/nd period extracted buildings
- : 2st/nd period extracted buildings



Approx. 1.6km

Approx. 1.6km

Pleiades Neo 2023/8 10:48
Incidence angle: 8.9°



Pleiades Neo 2025/1 10:48
Incidence angle: 7.6°

Satellite Image + Extracted Buildings (Enlarged view)

- : 1Extracted Buildings
- : 2Extracted Buildings
- : New construction
- : Demolished



Pleiades Neo 2023/08/04



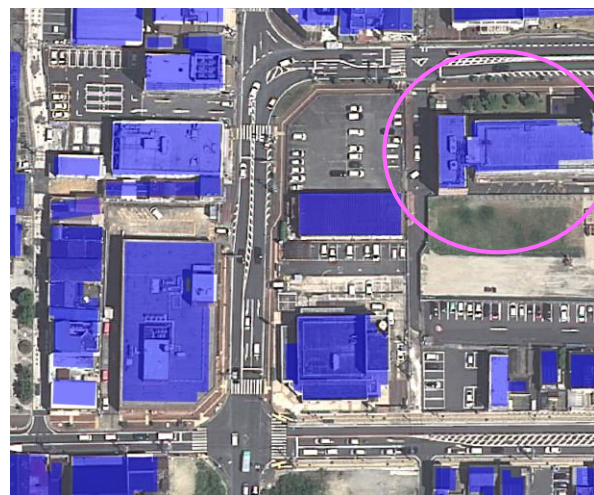
Pleiades Neo 2025/01/18



Pleiades Neo 2023/08/04



Pleiades Neo 2025/01/18



Pleiades Neo 2023/08/04+Extracted Buildings



Pleiades Neo 2025/01/18 +Extracted Buildings



Pleiades Neo 2023/08/04 +Extracted Buildings



Pleiades Neo 2025/01/18 +Extracted Buildings

Discussions –demonstration with municipal staff-

◆Unet-based model improves building extraction



Input Image



FY2023 result



FY2024: New algorithm

Built a general-purpose model with multi-region / multi-period training datasets

◆Improved building-change extraction accuracy

FY2023

Class	Precision	Recall
New	0.807	0.703
Demolished	0.729	0.700

FY2024

Class	Precision	Recall
New	0.948	0.901
Demolished	0.939	0.940

◆Public consultation from municipalities

- *Accuracy 90%+ is well-received.* Some demolished-related *detection-class coverage increase* would be even better
- *Valuable for streamlining interpretation work and for documenting evidence of change-interpretation locations*

additional tools for practical implementation

◆Development of visual interpretation tool



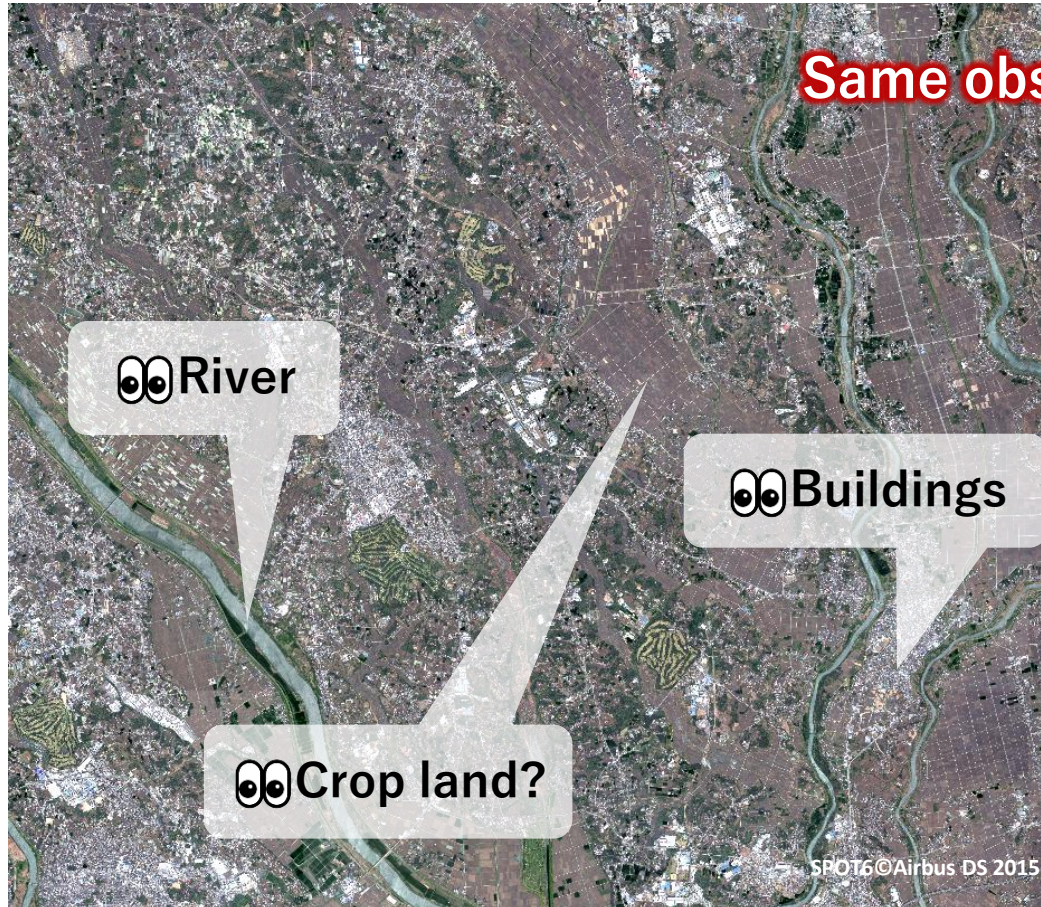
conventional workflow vs. workload
⇒50% reduction!

Multi-Source Change Detection for disaster analysis

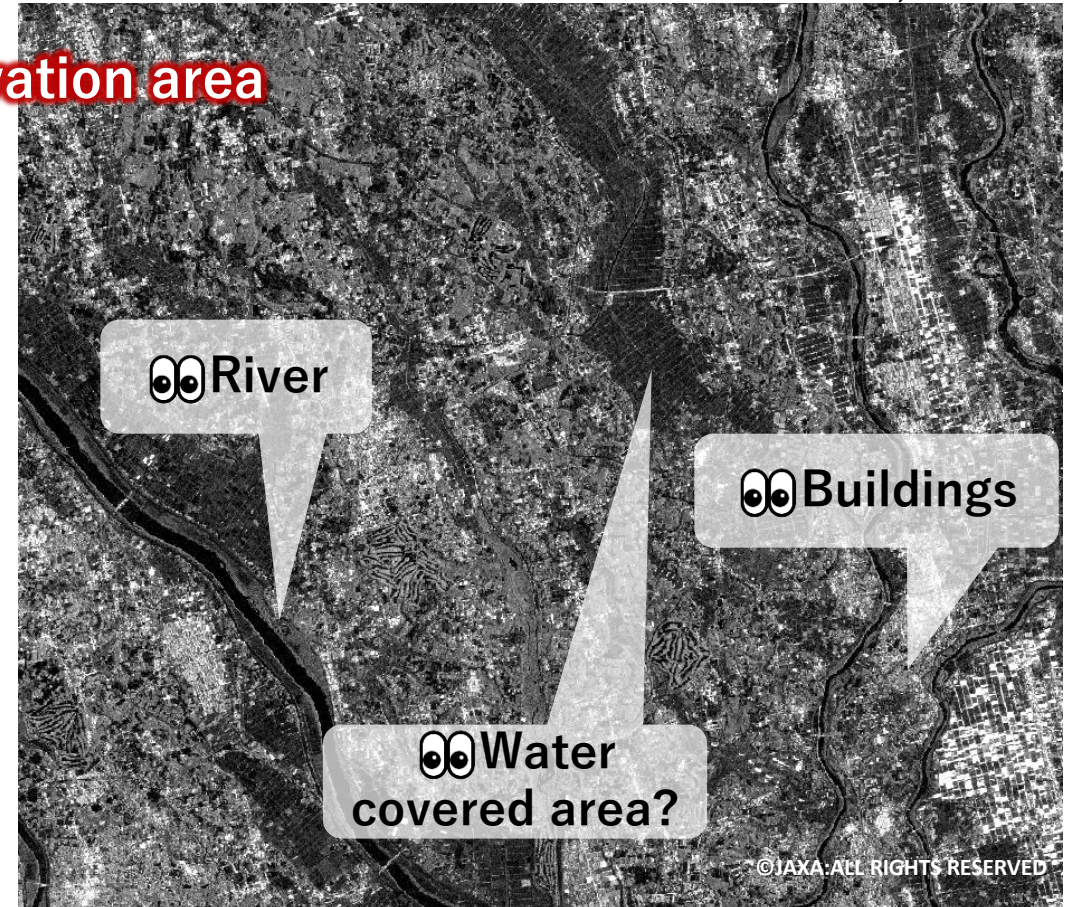
Differences in optical and SAR observation

- It is difficult to interpret and identify features between Optical and SAR image.

Optical data observed in April
(Passive: observe reflectance from land-surface)



SAR data observed in July
(Active: transmit microwave and observation its reflectance from land-surface)



Multi-Source Change Detection - Integration of Optical and SAR sensors -

Conventional remote sensing data analysis.

- ✓ Use of sensors with same specifications is common.
- ✓ Waiting time until next observation is long.

Our proposed approach

- ✓ Multi-Source data fusion.
- ✓ Allows for a shorter waiting time until the next observation.

Multi-sensor integration	Observation frequency	Observation condition (Cloud, night-time)	Features
Optical × Optical			👍 Aerial photographs can also be integrated.
SAR × SAR			👍 Available nighttime observation ✗ Difficulty in interpretation
Optical × SAR			👍 Improving disaster response capability ✗ Difficulty in analysis

Multi-source utilization — Optical data and SAR data —

Optical data
Easy to understand
Cannot capture under bad weather conditions
Enables change detection through image analysis
Accuracy drops due to different observation conditions before/after the event

Opposing



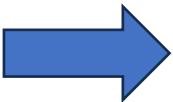
Common



SAR data
Difficult to understand
All-weather observation Capability
Enables change detection through image analysis
Accuracy drops due to different observation conditions before/after the event

Red text: Advantage
Blue text: Disadvantage

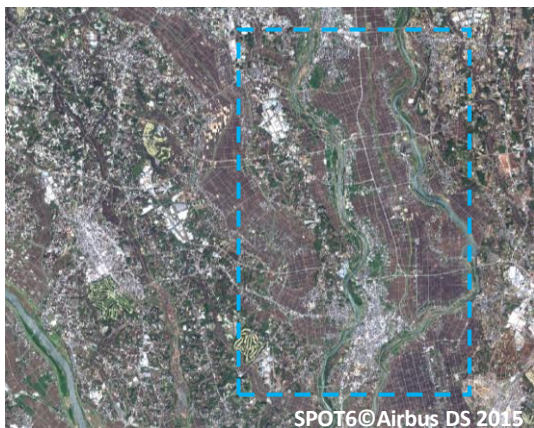
- Leverages the advantages of each sensor
- Resolves common issues via technical approach



Establishes a highly responsive observation system fusing Optical and SAR data

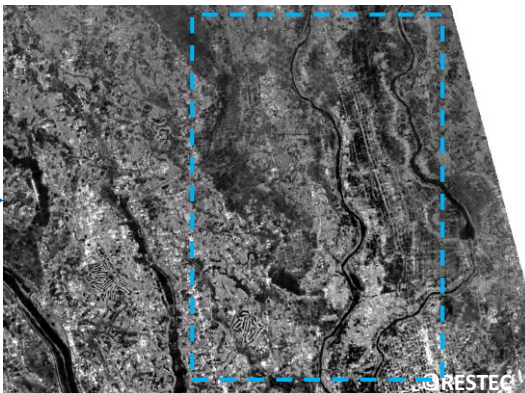
Multi-Source Utilization — Inundation Area Extraction

Change detection between optical (pre-disaster) and SAR (post-disaster) image ~2015 Kanto-Tohoku Heavy Rain (Joso Flood) ~



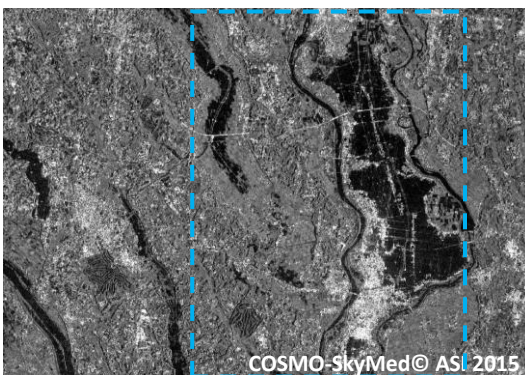
【Pre-disaster: 2015/4/15】

Optical image (SPOT-6)



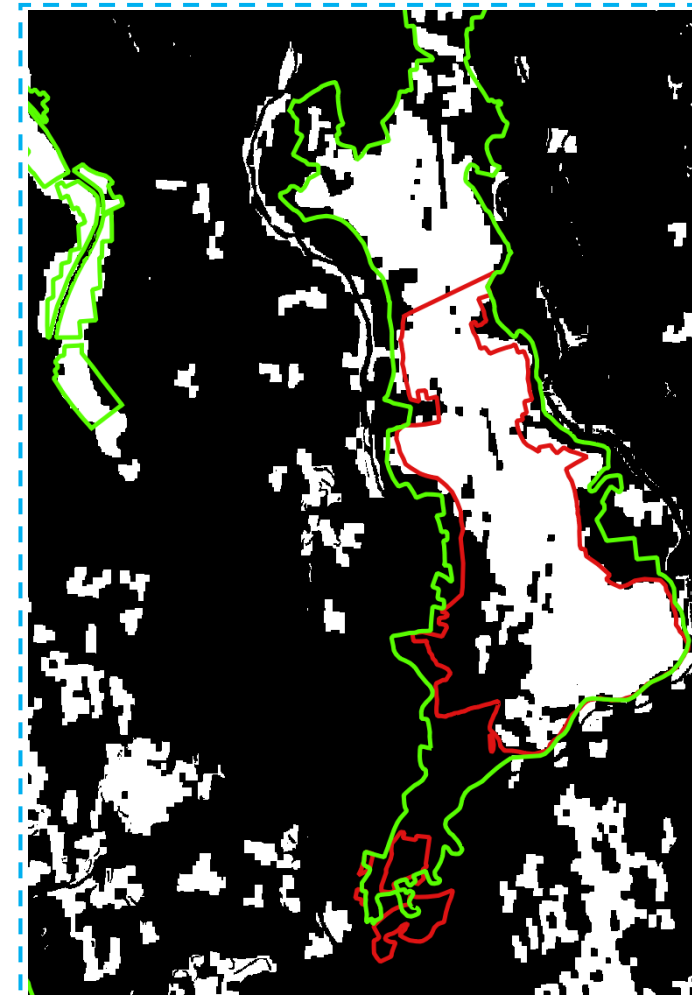
【Pre-disaster】

Pseudo-SAR image



【Post-disaster: 2015/9/11 17:12】

SAR image (COSMO-SkyMed)



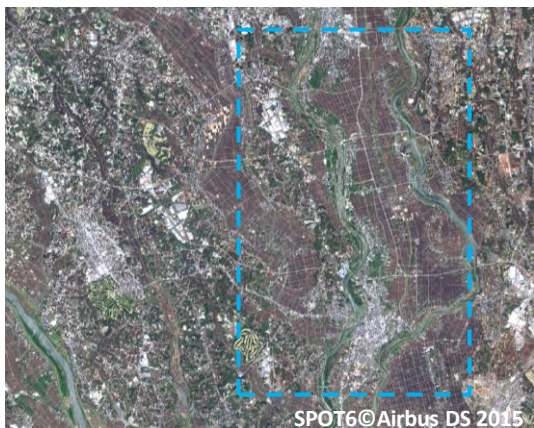
- White pixels: Estimated inundation area
- GSI-interpreted inundation area

□ 9/11 11:00
□ 9/12 15:30

- ✓ Enables change detection between Optical and SAR data by **Generating pseudo-SAR image from pre-disaster optical image based on AI inference**
- ✓ Replacing immediate post-disaster optical observation with SAR
- ✓ Boosting **responsiveness**
(= **no need to wait for clear skies**)

Multi-Source Utilization — Inundation Area Extraction

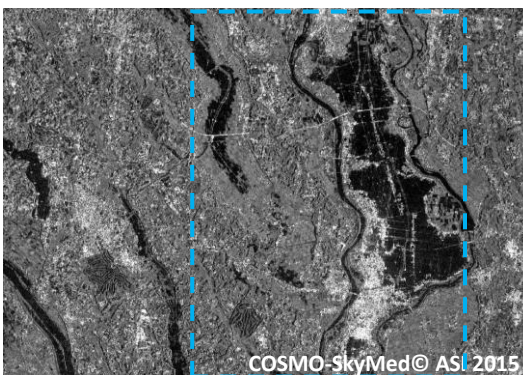
Change detection between optical (pre-disaster) and SAR (post-disaster) image ~2015 Kanto-Tohoku Heavy Rain (Joso Flood) ~



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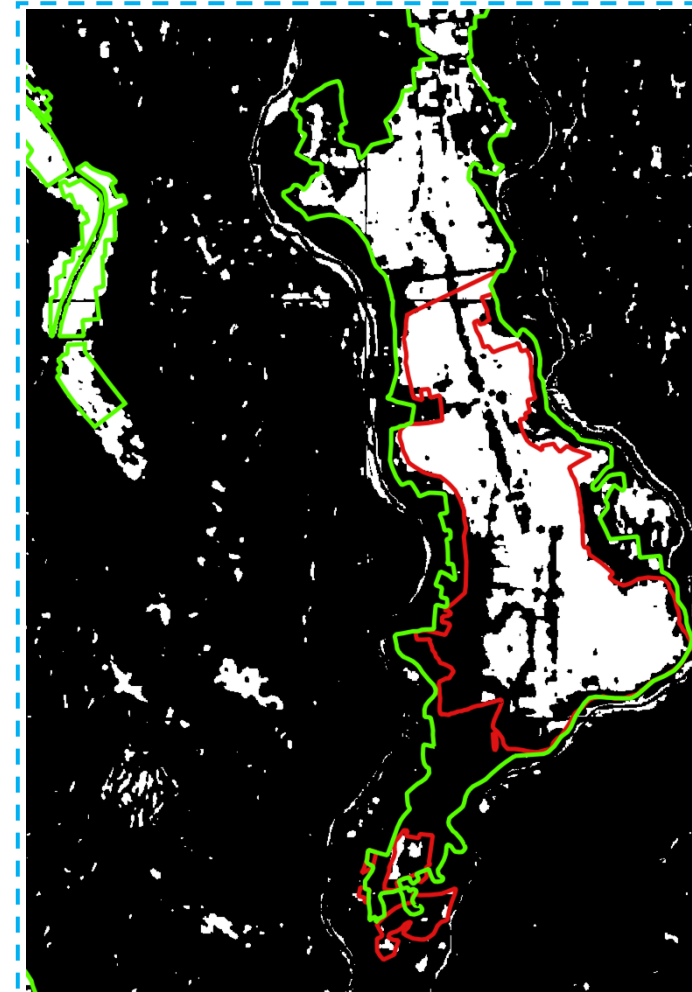
Optical image (SPOT-6)

- ✓ Develop an AI framework for direct comparison between Optical and SAR images
- ✓ Replacing immediate post-disaster optical observation with SAR
- ✓ Boosting responsiveness (= no need to wait for clear skies)



【Post-disaster: 2015/9/11 17:12】

SAR image (COSMO-SkyMed)



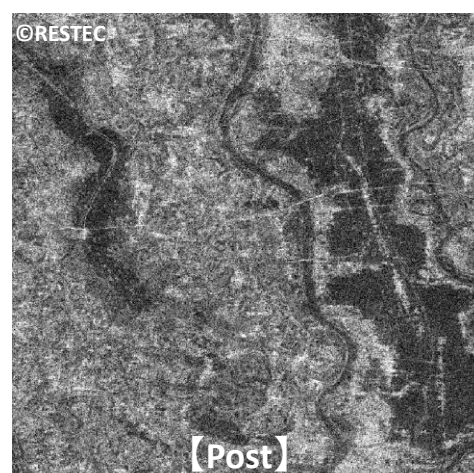
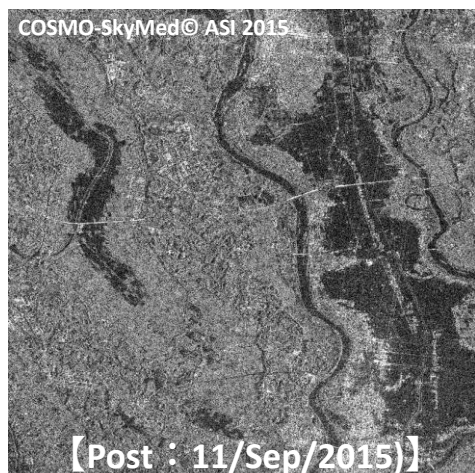
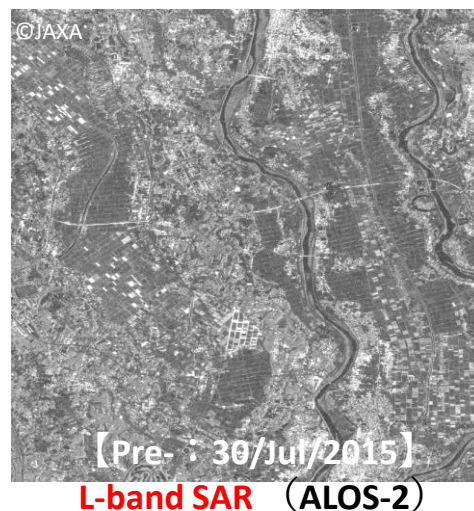
- White pixels: Estimated inundation area
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□ 9/11 11:00
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Multi-Source Utilization — Inundation Area Extraction —

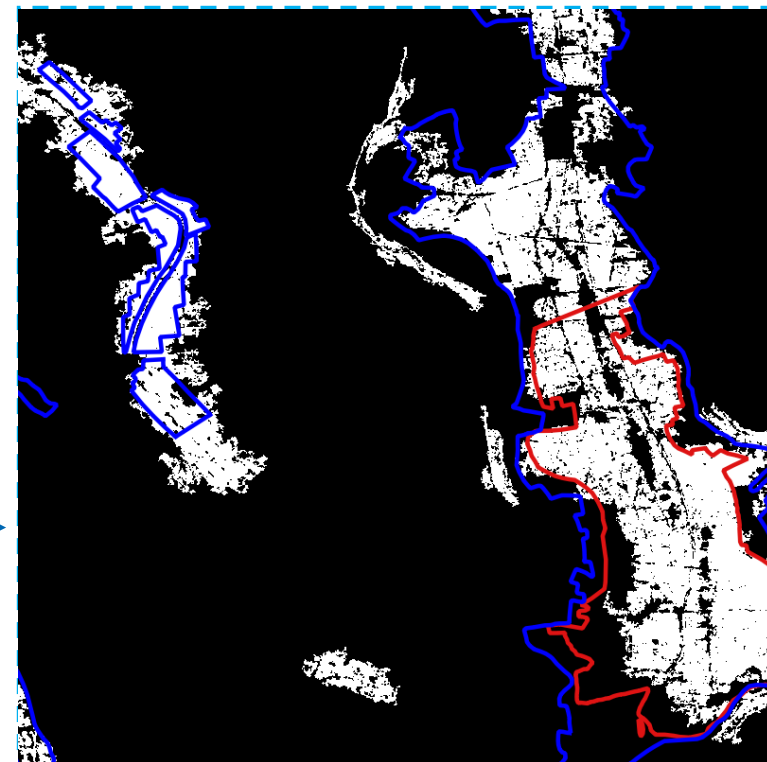
Change detection between different wavelength SAR images ~2015 Kanto-Tohoku Heavy Rain ~

- ✓ Enables change detection between X- and L-band SAR data **by generating pseudo L-band SAR image from post-disaster X-band SAR image based on AI inference**
- ✓ Enables immediate post-disaster SAR image analysis at any wavelength
- ✓ Boosting responsiveness (= no need to wait for clear skies)



X-band SAR (COSMO-SkyMed)

Pseudo-L-band SAR



Summary

Contribution to municipal DX = Solutions offered via Sat. image & AI

- ✓ Contribution to operational efficiency amid labor shortages
 - Labor-intensive tasks are reduced / automated by AI
 - Possibility of reducing cost (interpretation of aerial photos)

Disaster damage assessment via multi-source analysis

- ✓ Builds an AI model for direct comparison between Optical and SAR images analysis is possible regardless of weather
- ✓ Establishes wavelength-independent SAR analysis for flood disaster
 - ➔ Enables all-weather analysis system
 - ➔ Improved responsiveness in disasters
- ✓ Can extract diverse damage types including buildings, farmland, inundation



RESTEC

Sense Your Earth

 tanahashi_ren@restec.jp

 [@restec3426](#)

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Satellite Image + Extracted Buildings (Zoom Sample)

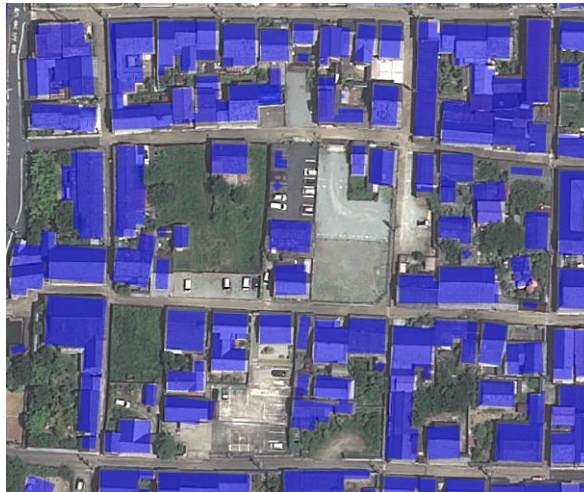
■: 1Extracted Buildings
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Pleiades Neo 2023/08/04



Pleiades Neo 2025/01/18



Pleiades Neo 2023/08/04+Extracted Buildings



Pleiades Neo 2025/01/18 +Extracted Buildings